Assigment 8

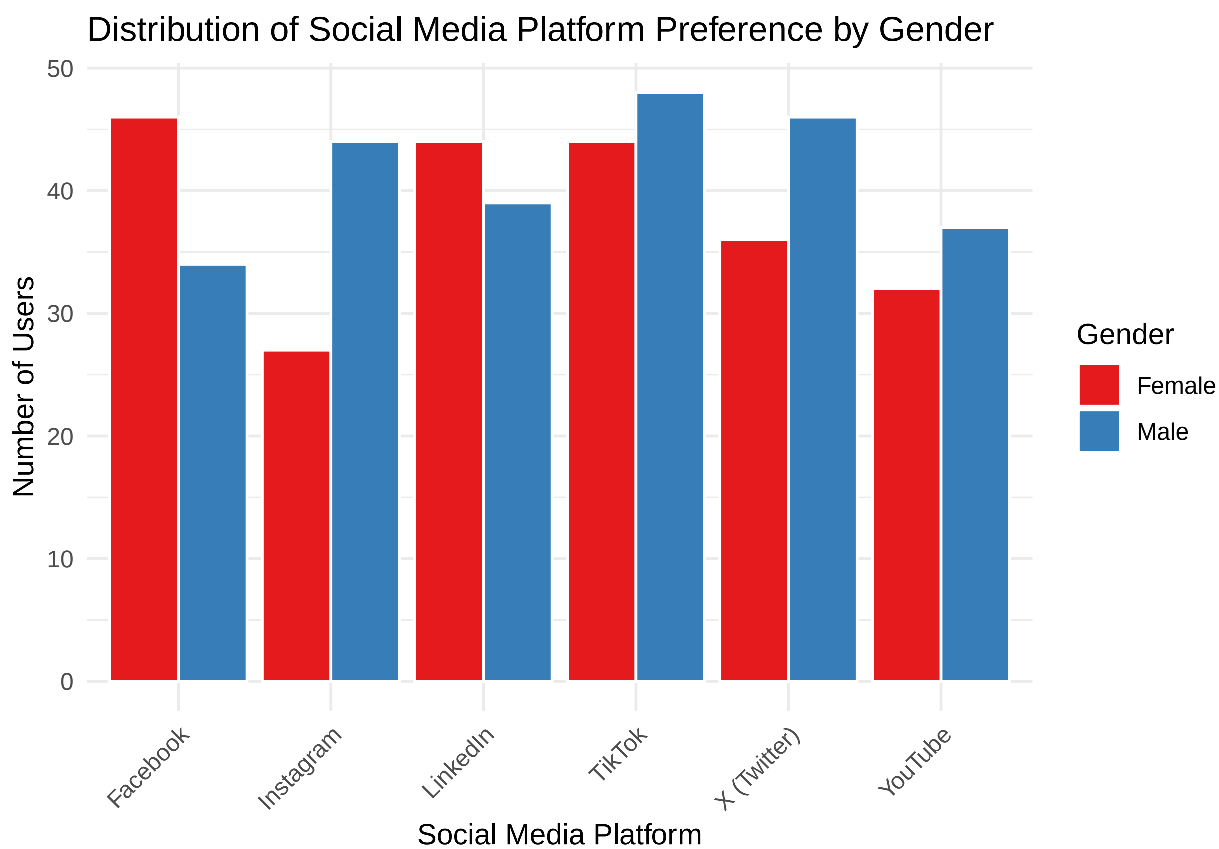
2025-11-29

```r
veri <- read.csv("Mental_Health_and_Social_Media_Balance_Dataset.csv")
```

```r
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

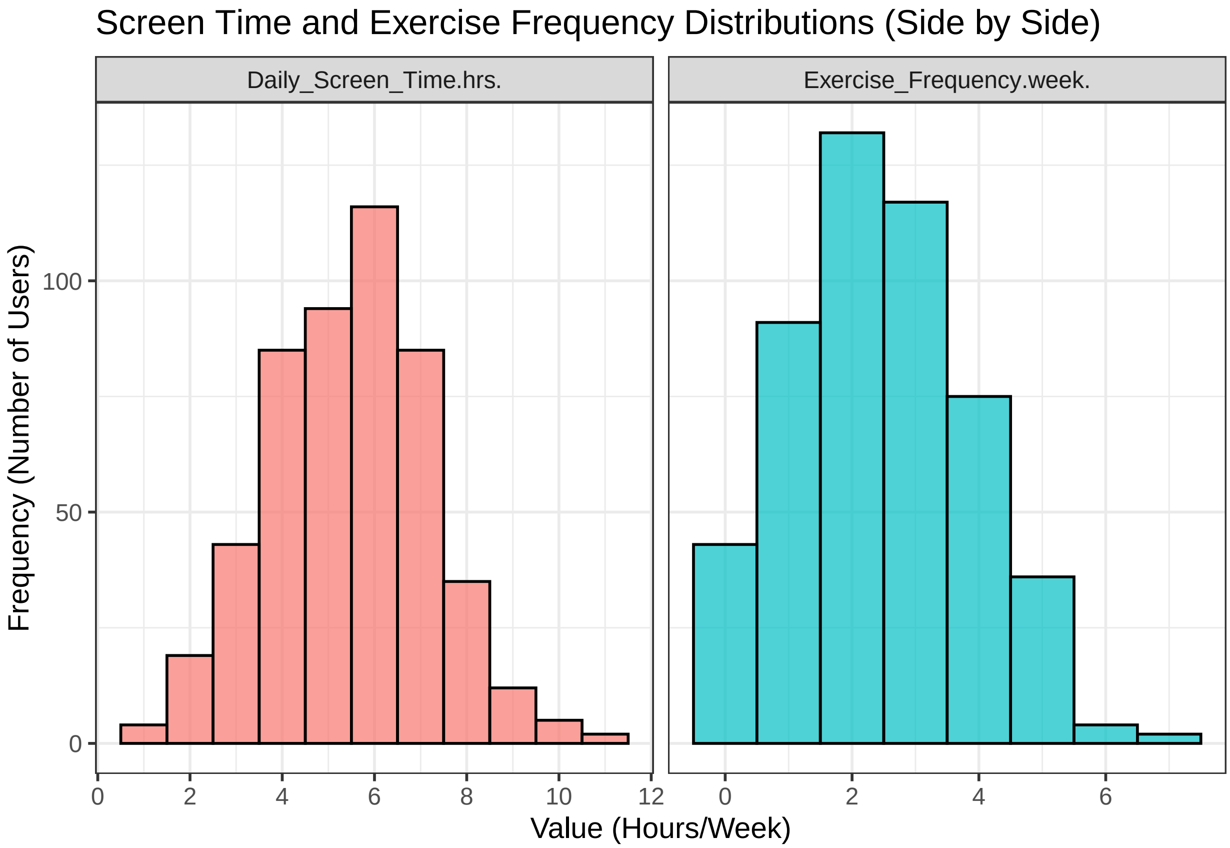
```r
veri_temiz <- veri %>%
  filter(Gender %in% c("Male", "Female"))

ggplot(veri_temiz, aes(x = Social_Media_Platform, fill = Gender)) +

  geom_bar(position = "dodge", color = "white") +


  labs(
    title = "Distribution of Social Media Platform Preference by Gender",
 x = "Social Media Platform",
y = "Number of Users",
fill = "Gender"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_fill_brewer(palette = "Set1")
```



1. Women Prefer Facebook, Men Prefer TikTok.
2. Female Participation is Lowest on Instagram.
3. TikTok and Facebook Are the Most Commonly Used Platforms.
4. LinkedIn Has the Closest Balanced Distribution.

```
library(ggplot2)
library(dplyr)
library(tidyr)

karsilastirma_verisi2 <- veri %>%
  select(Daily_Screen_Time.hrs., Exercise_Frequency.week.) %>%
  tidyr::gather(key = "Değişken", value = "Sayısal_Değer")

ggplot(karsilastirma_verisi2, aes(x = Sayısal_Değer)) +

  geom_histogram(aes(fill = Değişken), binwidth = 1, color = "black", alpha = 0.7) +

  facet_wrap(~ Değişken, scales = "free_x", ncol = 2) +

  scale_x_continuous(breaks = scales::breaks_pretty()) +

  labs(
   title = "Screen Time and Exercise Frequency Distributions (Side by Side)",
x = "Value (Hours/Week)",
y = "Frequency (Number of Users)"
  ) +
```

```r
  theme_bw() +
  theme(legend.position = "none")
```



Screen Time is highly centralized around 6 hours showing a normal distribution.

Exercise Frequency is concentrated between 1 and 3 times per week .

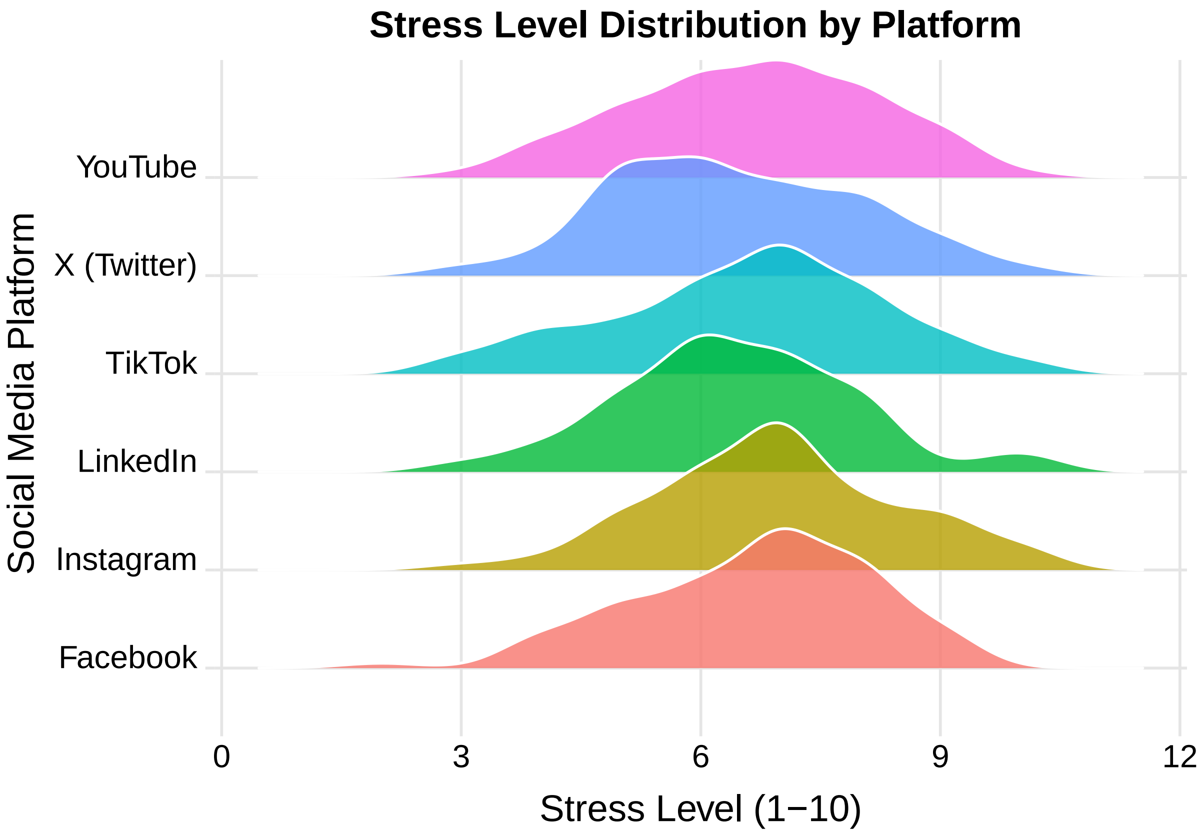
Few users report very high screen time (10+ hours) or very high exercise frequency (6+ times/week).

The screen time distribution is wider, suggesting a larger variance in daily hours compared to exercise frequency.

```r
library(dplyr)
library(ggridges)
library(ggplot2)

veri_sirali_ridge <- veri %>%
  group_by(Social_Media_Platform) %>%
  summarise(Ort_Stres = mean(Stress_Level.1.10., na.rm = TRUE), .groups = 'drop') %>%
  arrange(Ort_Stres) %>%
  mutate(Social_Media_Platform = factor(Social_Media_Platform, levels = Social_Media_Platform))

veri_ridge <- left_join(veri, veri_sirali_ridge, by = "Social_Media_Platform")

ggplot(veri_ridge, aes(x = Stress_Level.1.10., y = Social_Media_Platform, fill = Social_Media_Platform))
  geom_density_ridges(alpha = 0.8, color = "white", scale = 1.5) +

  labs(
    title = "Stress Level Distribution by Platform ",
```

```r
    x = "Stress Level (1-10)",
    y = "Social Media Platform"
  ) +

  theme_ridges() +
  theme(legend.position = "none",

        axis.title.x = element_text(hjust = 0.5),

        axis.title.y = element_text(hjust = 0.5),

        plot.title = element_text(hjust = 0.5)
  )
```

```
## Picking joint bandwidth of 0.511
```



Female users are most concentrated on Facebook, while Male users primarily dominate TikTok and X (Twitter).

Instagram records the lowest usage among Females, indicating a significant gender skew towards Males on that platform.

TikTok and Facebook are the most popular platforms overall, showing the highest combined user totals in the sample.

LinkedIn maintains the most balanced gender distribution, with Male and Female user counts being nearly identical.

A strong Male majority utilizes X (Twitter), suggesting higher male engagement with the platform's content.